Imperial-Cambridge Math Contest 2018 Rd 1 — P2/6

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(i) Show that if $(k+1)$ integers are chosen from $\{1, 2, \ldots 2k+1\}$, then among the chosen integers there are always two that are coprime.

(ii) Let $A = \{1, 2, \ldots, n\}$. Prove that if $n > 11$ then there is a bijective map $f : A \to A$ with the property that for every $a \in A$, exactly one of $f(f(f(f(a)))) = a$ and $f(f(f(f(f(a))))) = a$ holds.

Solution

(i) Let $1 \leq a \leq 2k$ be an integer. If both a and $(a+1)$ are chosen then we are done since a and $(a+1)$ are coprime.

Thus suppose we don't select two consecutive integers. There are exactly k even and $(k+1)$ odd numbers among $\{1, 2, \ldots, 2k+1\}$. If we chose the k even numbers, then we would have to pick one odd number which would create a pair of consecutive numbers. Thus the only way to pick $(k+1)$ non-consecutive numbers is to choose all of the odd numbers. However now since 1 was chosen and every number is coprime with 1, there does infact exist two coprime numbers among the chosen integers.

(ii) We examine two cases.

Case 1: n is even. Then define f like so,

$$f(a) = \begin{cases} a+1 & \text{if } a \text{ is odd,} \\ a-1 & \text{if } a \text{ is even.} \end{cases}$$

For odd a we have $f(f(a)) = a$ and so $f(f(f(f(a)))) = a$, whilst $f(f(f(f(f(a))))) = f(a) = a+1$. Similarly for even a we have $f(f(a)) = a$ and so $f(f(f(f(a)))) = a$ whilst $f(f(f(f(f(a))))) = f(a) = a-1$, as desired.

Case 2: n is odd. Then define f like so,

$$f(n-4) = n-3, \quad f(n-3) = n-2, \quad f(n-2) = n-1, \quad f(n-1) = n, \quad f(n) = n-4$$

and for $1 \leq a < n-4$,

$$f(a) = \begin{cases} a+1 & \text{if } a \text{ is odd,} \\ a-1 & \text{if } a \text{ is even.} \end{cases}$$

By the same arguement as in case 1, clearly the desired property is true for $1 \leq a < n-4$. But now for each $n-4 \leq a \leq n$ it can easily be verified that only $f(f(f(f(f(a))))) = a$ holds. ■